

Patrick Smith

Infrastructure and integration engineer with 13 years building production systems across Linux/AWS, APIs, databases, and customer support



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Summary

For thirteen years I built and maintained production systems from the ground up—architecting APIs that processed thousands of requests daily, deploying Linux infrastructure across AWS and Azure, optimizing MySQL databases, and providing technical support to hundreds of users. I combined technical depth (debugging with strace/tcpdump/gdb, performance optimization, security incident response) with operational breadth (customer support, documentation, after-hours on-call).

I'm strongest in the Linux/AWS/MySQL ecosystem with deep expertise in API integration (Stripe, PayPal, Twilio), REST/SOAP service design, database optimization, and systematic troubleshooting. I've recently expanded into AI/ML agent system design, building multi-agent workflows with deterministic simulation systems and automated code review pipelines.

What drives me is building systems that work reliably under production pressure, debugging failures methodically when they don't, and helping people succeed whether they're customers stuck on a problem or teams collaborating on a solution.

Experience

Founding Engineer

May 2011 – Sept 2024

Home Remodel Buddy, San Francisco, CA

- Served as sole technical support engineer for 13 years managing ~740 contractors and advertising partners, providing end-to-end case management from triage through root cause analysis to resolution, including after-hours support (nights, weekends, holidays).
- Built Cerberus API gateway (PHP) processing several thousand requests/day with sub-250ms response times, implementing two-phase validation (ping/post), custom HTTP headers for detailed error reporting, and rate limiting via rolling request windows.
- Designed Hercules wholesale lead distribution system with scoring algorithm balancing historical return rates (2-25%), pricing, and load distribution across partners generating \$25K-\$250K monthly revenue per partner.
- Built Mercury customer portal as PHP REST API consumed by Knockout.js SPA serving ~740 users with 30-second polling, rate limiting, and dual-mode authentication (PHP sessions + API keys for direct integrations).
- Implemented Autodial IVR system using Twilio Voice API and PHP, processing hundreds of double opt-in validation calls daily with outcome tracking (confirmed/voicemail/bad number).
- Integrated Stripe and PayPal payment processors with webhook event handling, idempotency keys for safe retries, and OAuth authentication flows, managing production payment processing for 13 years.
- Integrated third-party REST/SOAP APIs (WhitePages, Spokeo, BeenVerified, Facebook, Twilio) with async batch processing, retry logic, and error handling for lead validation and communication systems.
- Optimized MySQL database queries from $O(n^2)$ haversine calculations to bounding box pre-computation using EXPLAIN analysis and spatial indexing, reducing geographic batch processing from hours to minutes across ~5M row database.
- Used MySQL EXPLAIN to identify missing indexes and N+1 query patterns, adding targeted indexes to improve query performance and reduce database load.

Skills

- Programming
- Python

PHP

Bash

JavaScript

Lua
- Infrastructure & Cloud
- AWS

Docker

Git

Azure

Linux

CI/CD
- Networking
- VPC

DNS

VPN
- Databases
- MySQL

Memcached

Redis
- APIs & Monitoring
- REST

Grafana

SOAP

Other

Multi-Agent AI System Design

2025

Gridiron Dynasty (Open Source)

Architected collaborative AI agent system using Claude Code achieving 84% performance improvement (720s → 75s bootstrap) and 90% memory reduction (2.27GB → 224MB) through systematic optimization, parallel processing, and config caching. Designed specialized agent roles (Director, Architect, Engineer, Reviewer) with automated code review pipelines and comprehensive test coverage (83+ test cases).

Data Science

July 2025

HarvardX

- Designed MySQL database schemas for operational and analytical workloads, including normalization, indexing strategies, and query optimization.
- Wrote complex SQL queries with joins, subqueries, window functions, and aggregations to transform raw data into business intelligence.
- Debugged production incidents using Brendan Gregg's 60-second Linux performance methodology (uptime, dmesg, vmstat, mpstat, pidstat, iostat, sar, top) and advanced tools including strace for system call tracing, tcpdump for network inspection, lsof for file descriptor analysis, and gdb for process debugging.
- Responded to WordPress security compromise by dumping malicious RAT from process memory via /proc/{pid}/mem, reverse-engineering obfuscated PHP (base64/eval chains), and completing full restoration in 4-5 hours with zero data loss.
- Debugged fail2ban production outage that blocked nginx-to-apache internal communication, identified issue using iptables -L, manually removed DROP rule to restore service, then whitelisted internal IPs to prevent recurrence.
- Responded to AWS Route 53 DNS outage affecting production infrastructure, coordinating with AWS support and implementing temporary DNS failover procedures.
- Deployed and managed Linux infrastructure across AWS (EC2, RDS, S3, CloudWatch, Route 53) and Azure, including VPCs, VPNs, Docker containerization, and automated deployment workflows.
- Built custom blue-green deployment system using Git pulls to versioned directories with atomic symlink switching, enabling instant rollbacks and zero-downtime deployments.
- Executed PHP 5.6 to 8.2 upgrade across four production systems (Cerberus, Hercules, Mercury, Autodial) in 2 weeks full-time with zero downtime, resolving deprecated functions and library incompatibilities when hosting provider forced migration.
- Implemented CI/CD pipelines using Git and automated deployment scripts to streamline releases and reduce deployment errors.
- Configured multi-region cloud deployments with DNS routing (Route 53, Azure DNS) for high availability and disaster recovery.
- Implemented monitoring using Grafana and Prometheus with custom PHP instrumentation, tracking server metrics (CPU, memory, I/O), API response times, error rates, and business metrics for proactive incident detection.
- Ensured PCI compliance for stored customer payment data, conducting security audits and implementing recommended improvements.
- Monitored IDS systems, investigated suspicious activity, and executed incident-response protocols for security events.
- Managed endpoint protection, antivirus coverage, and system hardening across all environments.
- Created runbooks and technical documentation covering API integrations, troubleshooting workflows, and system architecture with the philosophy that documentation should guide someone six months later when context is forgotten.
- Managed Google Ads campaigns and built analytics dashboards to track conversion rates, reduce acquisition costs, and optimize marketing spend.
- Built dashboards and analytics tools to track KPIs, campaign performance, and customer funnel metrics in real-time.
- Developed Monte Carlo simulations in Python to model probabilistic outcomes and optimize decision strategies, running 300K+ iterations to calculate expected values and statistical confidence intervals.
- Conducted chi-square goodness-of-fit testing to validate PRNG implementations, applying hypothesis testing and statistical inference to audit randomness quality.
- Built custom analytics dashboards using Python (matplotlib) and JavaScript to visualize business metrics, conversion funnels, and performance KPIs with real-time data updates.
- Engineered Python ETL pipelines to extract, transform, and load data from RESTful and SOAP APIs into MySQL databases for analytics and reporting.

Office Assistant III

Nov 2024 – Present

San Benito County, HHSA, Public Assistance, Hollister, CA

- Provided customer service to San Benito County residents navigating government benefits systems (CalFresh, Medi-Cal), demonstrating that effective support means

References

**Christopher
Levine, Director
of Eng**

Lyft

415-827-2097,
clevine@keyobjects.com

**Barry
Willingham,
CEO**

*Funnel
Social*

415-655-1844,
barry@swimmingpoolquotes.com

meeting people where they are and explaining complex systems clearly without making them feel inadequate.

Media Operations Engineer

Jan 2011 - May 2011

Quotify Technology Inc. (acquired by Telstra), San Francisco, CA

- Developed and deployed custom WordPress templates for multiple business units.
- Managed Google Ads campaigns and increased landing page conversion from 8% to 12%.

Partner

Feb 2012 - Jan 2017

Founder's Dojo, San Francisco, CA

- Mentored early-stage founders on building and troubleshooting minimum viable products.

Education

A.S Business/Economics

2005-2007

Gavilan College, Gilroy, CA

Computer Science Coursework

2003-2005

California Polytechnic University, San Luis Obispo, San Luis Obispo, CA

CS109x: Introduction to Data Science with Python

July, 2025

HarvardX (Online Learning Initiative of Harvard University), <https://courses.edx.org/certificates/af05411c855d4ab7aaf9c8d90b29bd09>

Machine learning with sklearn, data analysis with Pandas/NumPy, regression and classification models, cross-validation, regularization, hypothesis testing, and statistical modeling

PH125.3x: Data Science: Probability

July, 2025

HarvardX (Online Learning Initiative of Harvard University), <https://courses.edx.org/certificates/73a1329507e642a0977384e4344c650e>

Probability theory, random variables, Monte Carlo simulations, Central Limit Theorem, expected values, standard errors, statistical inference foundations, and sampling models

Master Rank - Linux Troubleshooting

2025

SadServers.com, <https://sadservers.com/u/patrick/>

Completed advanced Linux troubleshooting scenarios including system debugging, performance analysis, networking issues, service failures, and production incident simulation